

NISCHAL ACHARYA, PhD

Data Scientist • Machine Learning • Quantitative Research

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Data Scientist with 7+ years building forecasting and time-series models on large, noisy, high-dimensional data across financial, geospatial, and scientific domains. Expert in Python and SQL, distributed processing (Spark/PySpark), and end-to-end pipelines from ingestion and feature engineering to deployment, MLOps, and stakeholder communication. Experienced with cloud and Databricks-style workflows and actively building with generative AI and agents. Focused on scalable, business-oriented solutions. Marie-Curie Fellow; fluent in English and Spanish; based in Madrid.

CORE SKILLS

Languages & DB: Python (advanced), SQL, R, Bash • PostgreSQL, MySQL
ML / Statistics: Forecasting & time series (ARIMA, Prophet, XGBoost), scikit-learn, PyTorch, Keras, Random Forest, Gradient Boosting, SVM, Bayesian inference, MCMC, model optimization, regression, classification, clustering
Data Eng & Cloud: Spark / PySpark, Databricks, Azure, ETL & data pipelines, Pandas, NumPy, SciPy, REST APIs, Docker, Git/GitHub/GitLab, CI/CD, MLOps, Generative AI & agents
Visualisation / BI: Power BI, Matplotlib, Plotly, Tableau (working), Excel (advanced)
Languages spoken: English (fluent) • Spanish (fluent) • German (intermediate) • Nepali (native)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher & Data Analyst — CEFCA, Teruel, Spain Feb 2025 – Present

- Lead end-to-end analysis of multi-wavelength survey data on **100,000+ sources**, building reproducible Python pipelines for ingestion, cleaning, and statistical modelling.
- Apply classification and probabilistic models to identify rare ionisation signatures (LINER AGN); translate complex spectral data into evidence-based conclusions for international stakeholders.
- Publish in peer-reviewed journals and present findings to non-specialist audiences — sharpening data storytelling for technical and executive audiences alike.

Quantitative Developer & Researcher — QuantumScale, Munich (remote) Jun 2025 – Nov 2025

- Designed and back-tested CFD trading strategies on crypto and equities; delivered **68.5% win rate, 1.12 profit factor, 10.3% max drawdown** on BTCUSDT across 9,679 trades.
- Built a production Python application for real-time market monitoring, multi-exchange (Binance, Bitget) data ingestion, strategy back-testing, and risk management — deployed as a MetaTrader 5 portfolio.
- Engineered the full quant stack: data-feed handlers, strategy engine, equity-curve & Sharpe analytics, scalability tests, and a Qt-based dashboard.

Data Scientist — OmegaLambdaTec GmbH, Garching, Germany Jul – Sep 2022

- Built supervised ML models (Random Forest, SVM) on **EU Sentinel satellite imagery** for automated land-cover classification — production-ready outputs delivered to a commercial geospatial team.
- Implemented end-to-end pipelines (scikit-learn, pandas, rasterio) with version-controlled, reproducible workflows; contributed to code reviews and Agile sprint cycles.

Quantitative Data Analyst — LMU Munich, Germany Nov 2023 – Jan 2024

- Processed **terabyte-scale simulation outputs** (MAGNETICUM); built Python pipelines to extract, transform, and statistically characterise structural properties.

SELECTED PROJECTS

- Algorithmic Trading Platform** — multi-exchange backtesting & live-monitoring app with strategy library, parameter sweeps, and Sharpe/drawdown reporting (Python, Qt, REST/WebSocket).
- miniJPAS Faint X-ray AGN Pipeline** — end-to-end Bayesian inference pipeline filtering 100k+ sources into a high-quality scientific sample; *first-author publication in A&A*.
- CIGALE** (open-source Python module) — co-developed and released a reusable module for probabilistic SED modelling; version control, code review, and packaging best practices.

EDUCATION

PhD, Astronomy (Marie-Curie Fellow) — Donostia International Physics Center, San Sebastián 2020 – 2025
Focus: large-scale survey data, Bayesian modelling, statistical inference, Python pipelines.

MSc, Astronomy & Space Physics — University of Turku, Finland 2020
Thesis: statistical analysis of 200,000+ galaxies — feature engineering, hypothesis testing, MCMC.

BSc, Physics — St. Xavier’s College, Kathmandu 2015

PUBLICATIONS & TALKS

First- and co-author of peer-reviewed papers in *A&A* and *ApJ*; 7+ invited talks across Europe (Copenhagen, Munich, Reykjavik, Granada, Turku, Bologna). Full list: [ORCID 0000-0001-9738-3594](#).